

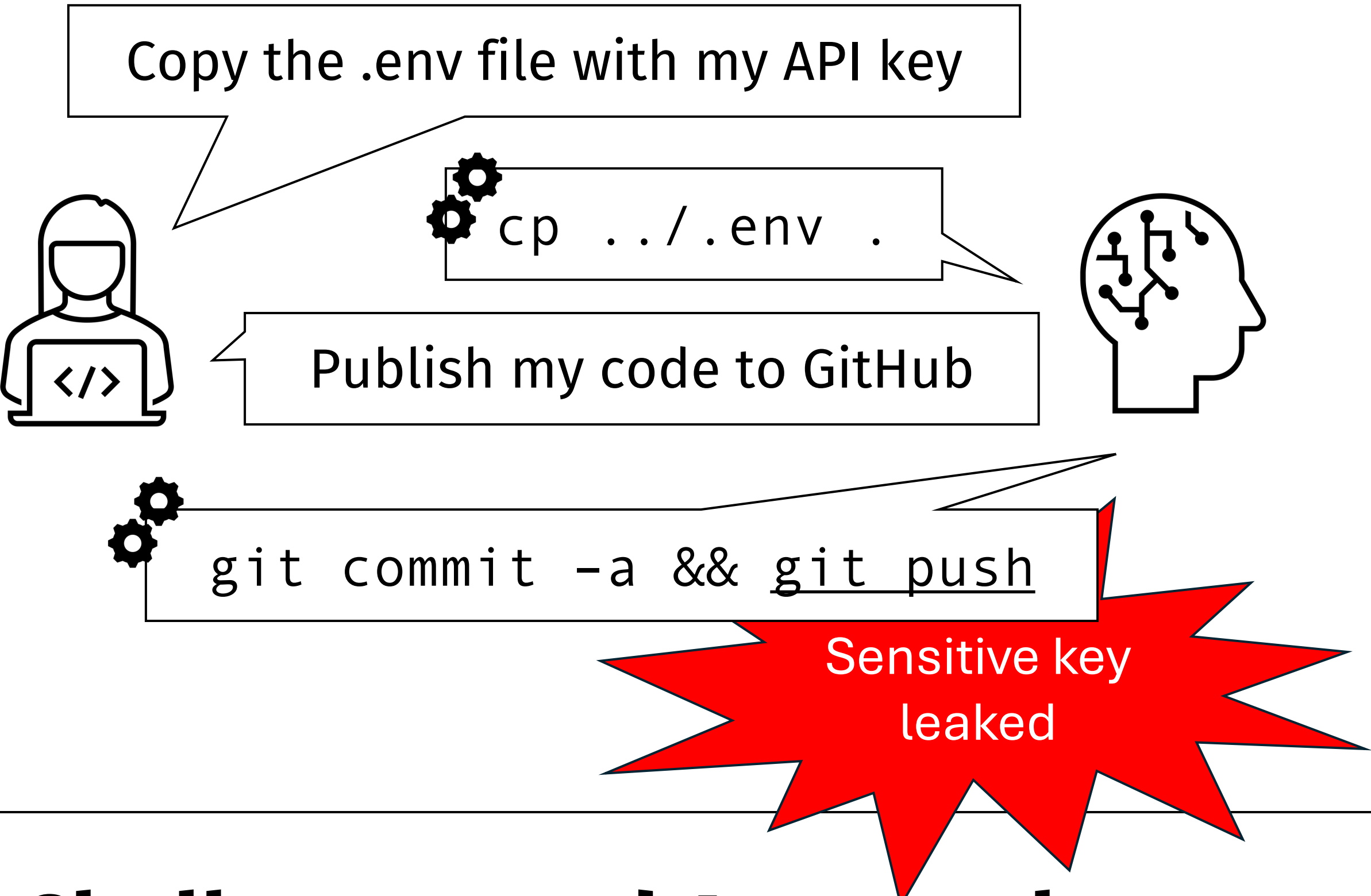


Rest Easy: Constraining Untrusted Agent Tool Calls with Haven

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Motivation



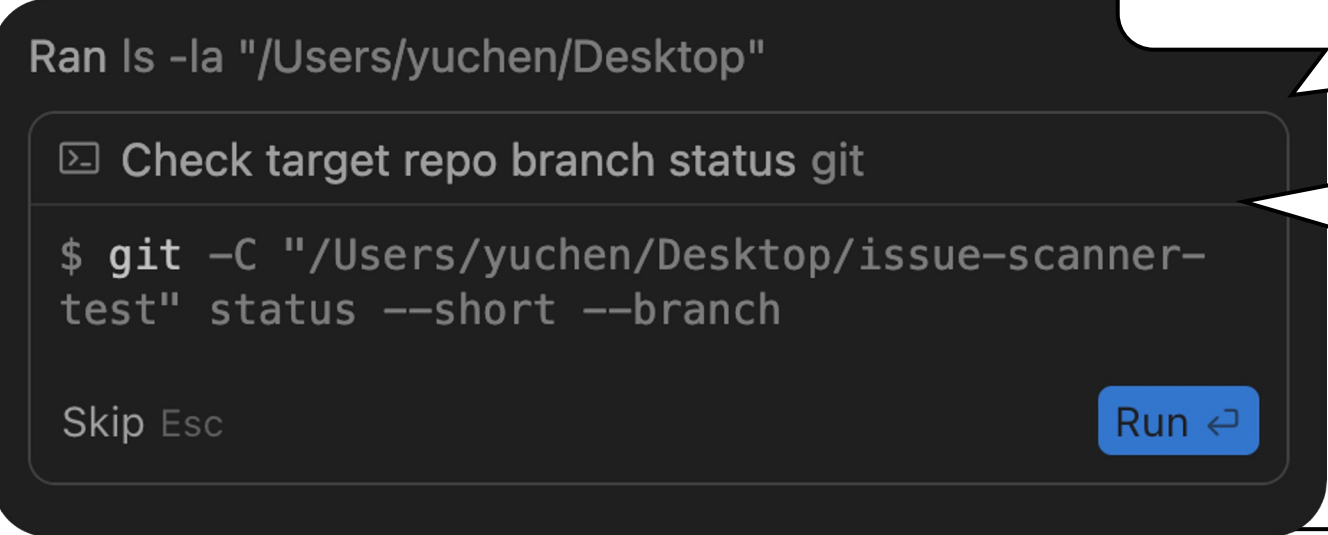
Other failure modes:

- Tool effects that are not disclosed in the description (e.g. logging, metrics collection)

In all cases a tool causes unintended or harmful *external effects*.

What providers do today

Permission Prompts



Users need to be experts

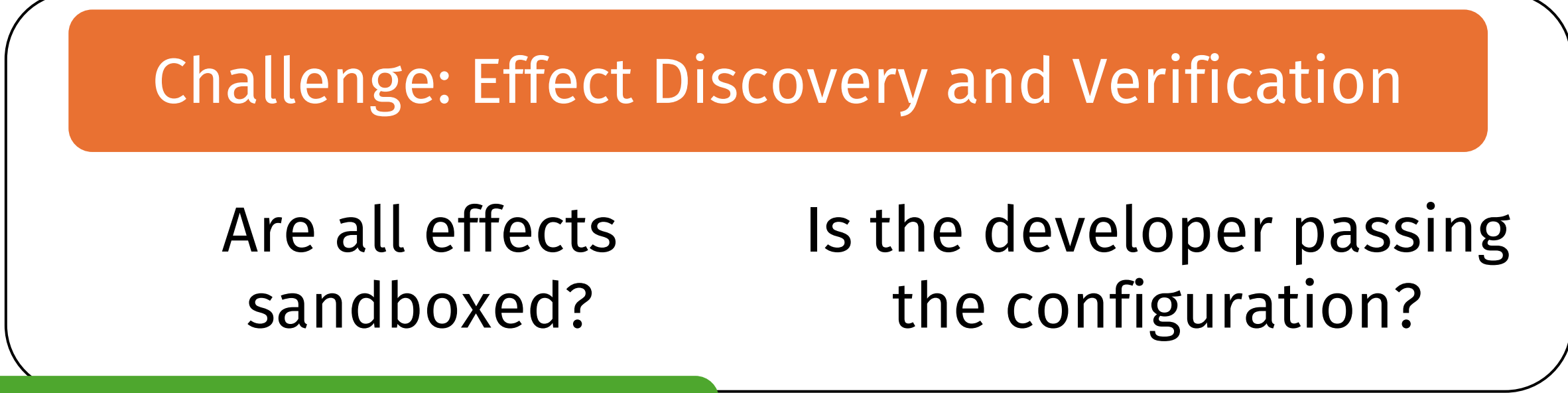
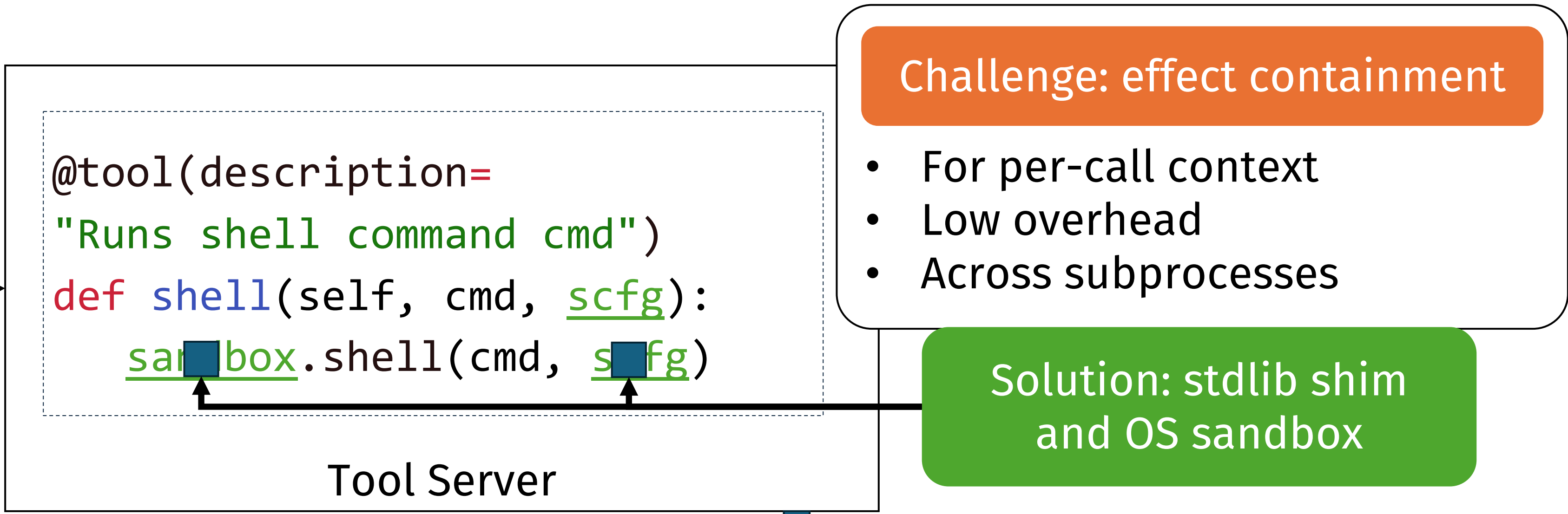
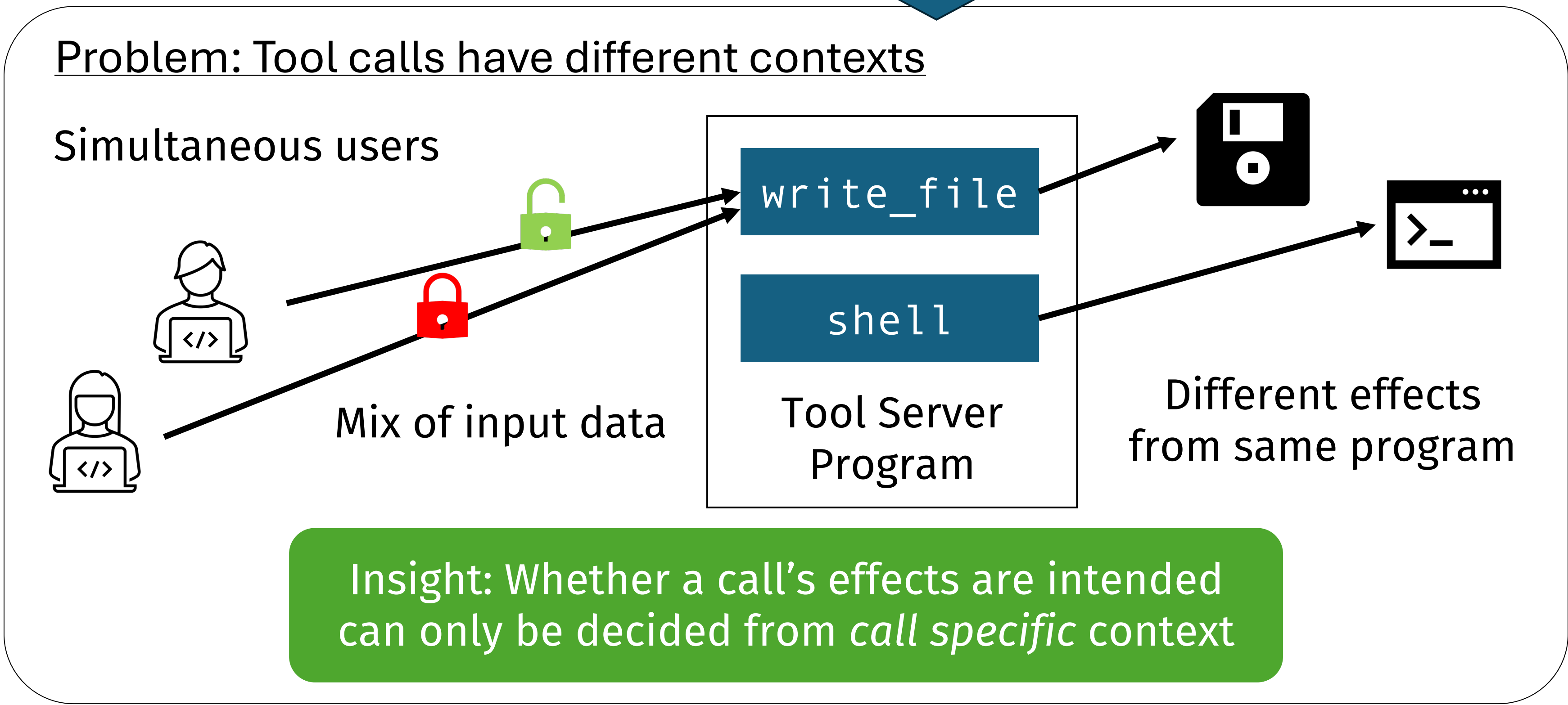
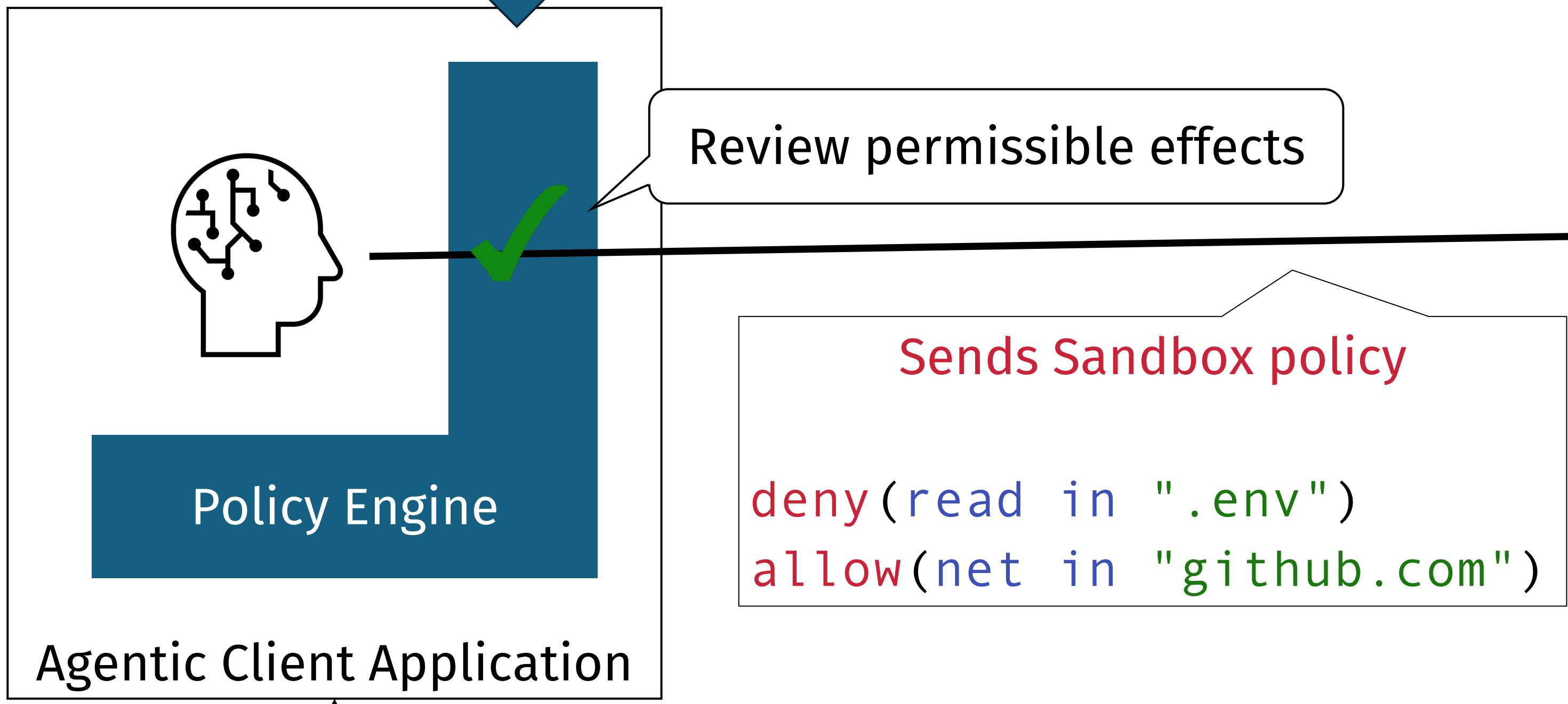
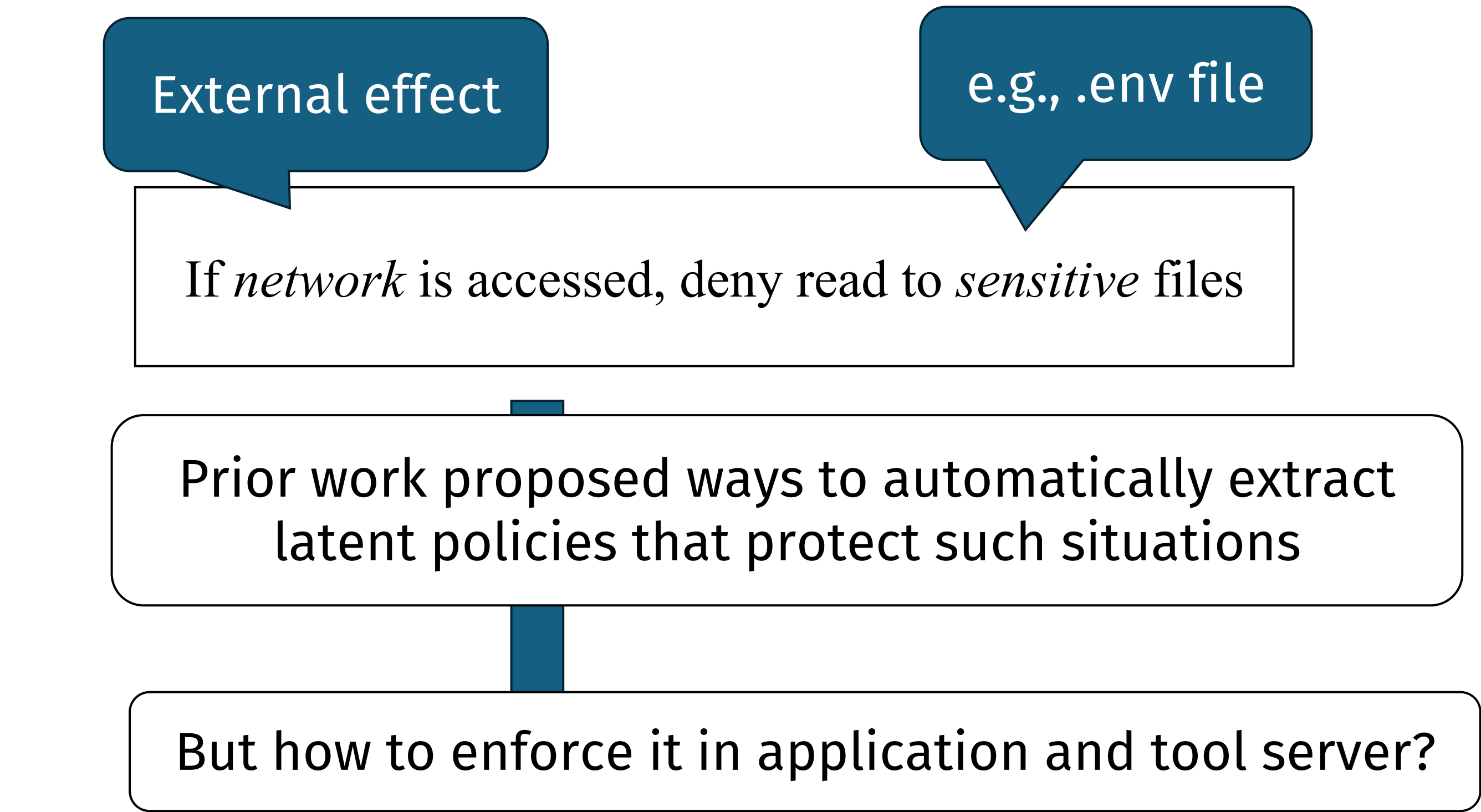
Causes Alert Fatigue

Global Sandboxes

Allow network egress
Give Claude network access to install packages and libraries in order to perform advanced data analysis, custom visualizations, and specialized file processing. Monitor chats closely as this comes with security risks.

Coarse. Either blocks everything or nothing

Challenges and Approach



Verification and discovery summary distributed to client

Results

Six interaction traces with Claude and one IDE MCP server.

Paired as benign and problematic variants to test utility and security.

Protection Mechanism	.env file		Sensitive Code		git push	
	Intended	Divergent	Intended	Divergent	Intended	Divergent
Unprotected	✓	✗	✓	✗	✓	✗
Global Sandbox	✗	✓	✓	✗	✓	✗
Static Effect Detection	✗	✓	✓	✓	✓	✓
Hybrid Approach	✓	✓	✓	✓	✓	✓